

You are a principal product engineer, enterprise banking solution architect, senior UX designer, and AI/ML engineer.

Your task is to DESIGN AND BUILD a complete, polished, demo-ready Proof-of-Concept named:

YELLOWSENSE CUSTOMER 360

Unified AI-Powered Customer Growth, Field Mobilization & Relationship Intelligence Platform

Company:

YellowSense Technologies

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0. PRIMARY OBJECTIVE

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Build a visually exceptional, realistic, interactive enterprise banking platform that unifies:

1. ZRT Field Visit & Business Mobilization
2. Relationship Manager (RM) Operations
3. Virtual Relationship Manager (VRM) Operations
4. Central Customer 360
5. Lead and Opportunity Management
6. Customer Meeting Intelligence
7. Query and Complaint Management
8. AI Decisioning
9. Management Analytics and MIS
10. Role-Based Access Control
11. Consent and Audit Visibility
12. Offline-first field workflow simulation

The application must be suitable for a live pitch to senior stakeholders of a potential bank.

This is NOT:

- a generic CRM
- a simple admin dashboard
- a collection of disconnected pages
- a static Figma-like mockup
- a toy chatbot
- a basic CRUD application

This IS:

- a coherent banking execution platform
- a premium enterprise product

- a working POC with realistic end-to-end journeys
- a role-based system built around one shared Customer 360
- an AI-assisted decision and execution layer

The central proposition is:

"One Customer. One Intelligence Layer. Every Team Acting Together."

The platform should demonstrate this lifecycle:

PROSPECT

- FIELD VISIT
- NEED DISCOVERY
- CUSTOMER 360 ENRICHMENT
- AI OPPORTUNITY DETECTION
- LEAD CREATION
- SMART ASSIGNMENT
- RM/VRM ENGAGEMENT
- MEETING
- FOLLOW-UP
- CONVERSION
- ONBOARDING
- CROSS-SELL
- SERVICE
- RETENTION
- CONTINUOUS LEARNING

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1. NON-NEGOTIABLE BUILD PRINCIPLES

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Follow these principles:

A. BUILD A WORKING PRODUCT

Every important button should perform a meaningful action.

B. BUILD END-TO-END JOURNEYS

Do not create disconnected screens.

C. USE REALISTIC BANKING DATA

Use fictional customers only, but realistic Indian banking terminology, values, workflows and business scenarios.

D. DESIGN FOR BANK EXECUTIVES

The product must look credible in a CIO/CDO/Business Head presentation.

E. AI MUST BE EXPLAINABLE

Never show unexplained AI scores.

F. HUMAN-IN-THE-LOOP

AI recommends; authorized employees decide.

G. NO FALSE COMPLIANCE CLAIMS

Do not display "100% RBI compliant" or "fully DPDP compliant."

Use:

"Designed to support applicable security, privacy, audit and regulatory control requirements."

H. NO RIP-AND-REPLACE POSITIONING

Existing CBS, LOS, cards and other systems remain systems of record.

I. POC-FIRST ARCHITECTURE

Keep the implementation simple enough to run locally while preserving clean enterprise boundaries.

J. NO OVERENGINEERING

Do not add Kafka, Kubernetes, Temporal, Databricks or microservices unless implemented meaningfully.

For the POC, simulate enterprise integrations behind adapters.

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2. BRANDING AND VISUAL IDENTITY

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Brand:

YellowSense Technologies

Product:

YellowSense Customer 360

Theme:

Premium light enterprise banking interface.

Color palette:

Background:

#FFF9ED

Surface:

#FFDF7

Primary Navy:
#16263A

Primary Yellow:
#FFD51F

Warm Gold:
#F4A623

Warm Orange:
#FF8A16

Primary Text:
#29313A

Muted Text:
#6B7076

Success:
#2F8467

Warning:
#D89A20

Danger:
#B74C44

Borders:
#E8DAAE

DESIGN LANGUAGE:

Use:

- warm ivory backgrounds
- premium white/cream cards
- deep navy typography
- yellow-orange accents
- subtle shadows
- generous whitespace
- 12px to 18px corner radius
- thin borders
- strong information hierarchy

- sophisticated charts
- realistic tables
- contextual side drawers
- command palettes
- hover states
- skeleton loading states
- meaningful empty states
- subtle transitions
- responsive layouts

VISUAL MOTIF:

Use concentric rings and orbital arcs inspired by the YellowSense logo.

Apply these motifs subtly to:

- Customer 360
- AI decision engine
- trust/security
- loading indicators
- hero backgrounds
- relationship visualizations

Do not overuse gradients.

Avoid:

- neon cyberpunk
- excessive glassmorphism
- purple AI gradients
- generic blue SaaS appearance
- giant empty cards
- emoji as product icons
- excessive rounded pills
- random decorative illustrations
- fake 3D visuals
- cluttered dashboards

Use Lucide icons or an equivalent professional icon system.

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3. RECOMMENDED TECH STACK

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Create a monorepo.

Frontend:

- Next.js 15+
- App Router
- TypeScript strict mode
- Tailwind CSS
- shadcn/ui
- Radix primitives
- Lucide icons
- Recharts
- TanStack Table
- TanStack Query
- React Hook Form
- Zod
- Zustand only where global client state is genuinely needed
- Framer Motion for restrained transitions

Backend:

- FastAPI
- Python 3.12+
- Pydantic v2
- SQLAlchemy 2
- Alembic
- PostgreSQL
- pgvector-ready schema
- Redis optional
- structured logging

POC mode:

- Docker Compose
- PostgreSQL container
- optional Redis container
- frontend container
- backend container

Testing:

Frontend:

- Vitest
- React Testing Library
- Playwright

Backend:

- pytest
- httpx test client

Code quality:

- ESLint
- Prettier
- Ruff
- mypy where practical

Do not use Firebase as the primary architecture.

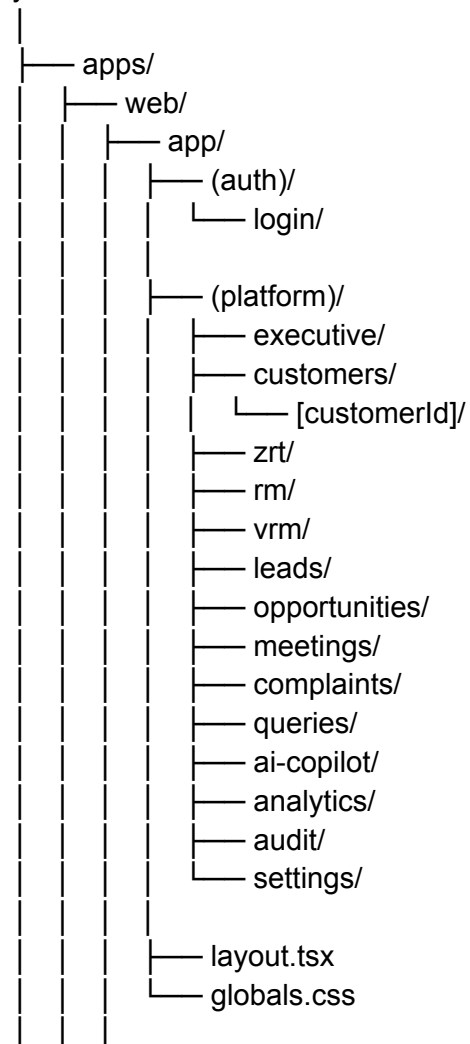
Do not put all business logic inside Next.js route handlers.

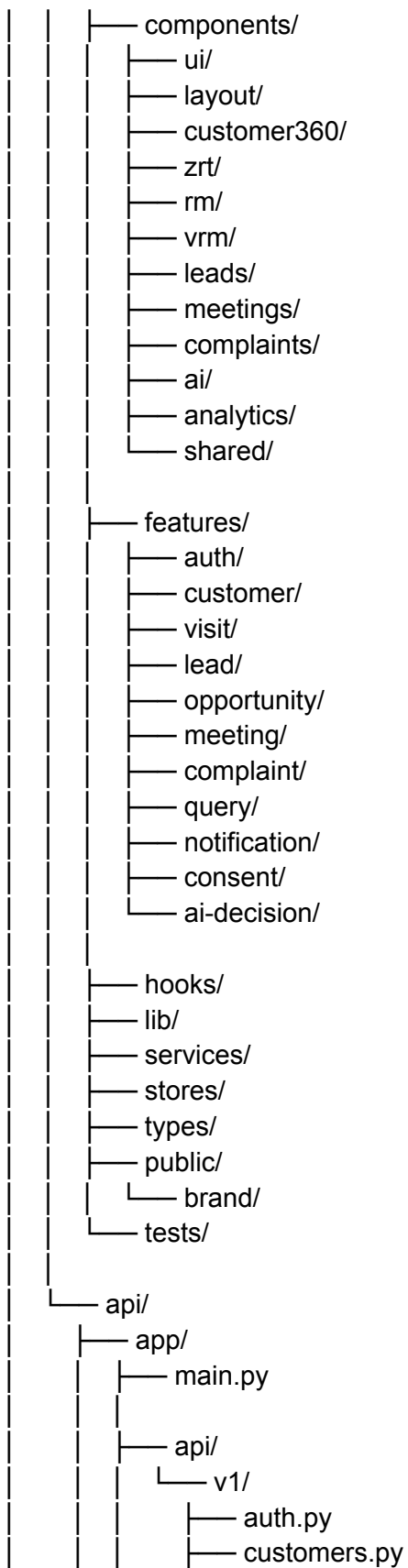
Do not put AI logic directly inside UI components.

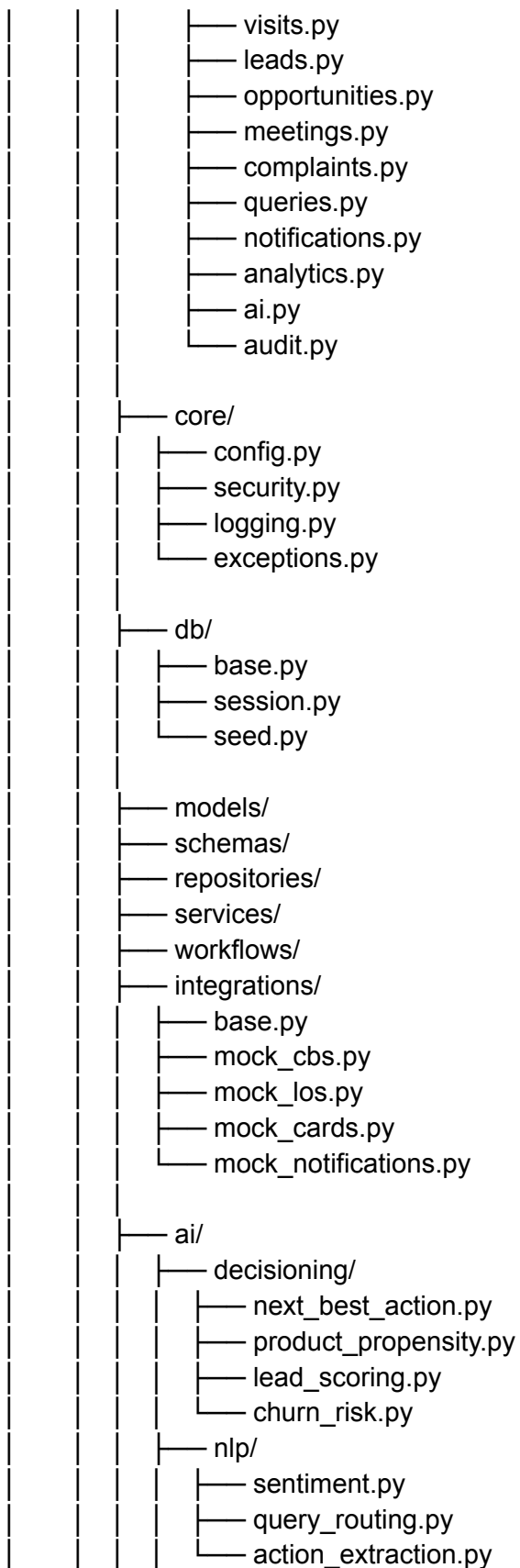
4. REQUIRED MONOREPO FOLDER STRUCTURE

Create exactly this clean structure, adjusting only when technically necessary:

yellowsense-customer360/







```

├── meeting_intelligence/
│   ├── copilot/
│   ├── explainability/
│   └── providers/
├── tests/
├── alembic/
└── pyproject.toml

packages/
├── design-tokens/
├── shared-types/
└── config/

infra/
├── docker/
├── nginx/
└── compose/

docs/
├── architecture/
├── api/
├── demo/
└── adr/

scripts/
├── seed_demo_data.py
├── reset_demo.py
└── smoke_test.py

.env.example
docker-compose.yml
Makefile
README.md
ARCHITECTURE.md

```

5. CORE DOMAIN MODEL

Implement realistic entities.

User

- id
- employee_id
- name
- email
- role
- branch_id
- region_id
- status

Role values:

- ZRT_OFFICER
- RM
- VRM
- BRANCH_MANAGER
- REGIONAL_MANAGER
- HEAD_OFFICE
- ADMIN

Customer

- id
- customer_number
- full_name
- customer_type
- segment
- lifecycle_stage
- mobile
- email
- city
- state
- branch_id
- assigned_rm_id
- assigned_vrm_id
- relationship_value
- relationship_tenure_months
- digital_engagement_score
- sentiment
- churn_risk
- lead_propensity
- created_at
- updated_at

CustomerProfile

- demographics
- occupation

- business_type
- annual_income
- annual_turnover
- employee_count
- preferred_language
- preferred_channel

Account

- account_number_masked
- account_type
- balance
- status
- opened_at

ProductHolding

- product_type
- product_name
- status
- value

Interaction

- channel
- interaction_type
- direction
- summary
- sentiment
- occurred_at
- employee_id

Visit

- customer_id
- zrt_officer_id
- purpose
- scheduled_at
- check_in_at
- check_out_at
- latitude
- longitude
- geo_verified
- notes
- status
- sync_status

NeedAssessment

- visit_id
- working_capital_need
- term_loan_need
- pos_qr_need
- salary_account_need
- insurance_need
- notes

Lead

- id
- customer_id
- source
- product
- potential_value
- stage
- owner_id
- conversion_probability
- priority
- next_action
- next_action_due_at
- created_at

Opportunity

- lead_id
- expected_value
- expected_close_date
- stage
- probability

Meeting

- customer_id
- employee_id
- scheduled_at
- purpose
- transcript
- summary
- sentiment
- action_items
- follow_up_date

Complaint

- customer_id
- category
- severity

- status
- assigned_team
- sla_due_at
- sentiment
- escalation_level

Query

- customer_id
- source_channel
- raw_text
- detected_intent
- urgency
- routed_team
- status

Consent

- customer_id
- channel
- purpose
- granted
- captured_at
- revoked_at

AIRecommendation

- customer_id
- recommendation_type
- recommendation
- confidence
- reason_codes
- model_name
- model_version
- generated_at
- employee_decision
- decision_at

AuditEvent

- actor_id
- action
- entity_type
- entity_id
- before_state
- after_state
- timestamp

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6. REQUIRED DEMO PERSONAS

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Create a role switcher for demo purposes.

Demo users:

1. Arjun Rao

Role: ZRT Officer

Region: Chennai Central

2. Priya Nair

Role: Relationship Manager

Specialization: MSME

3. Vikram Shah

Role: Virtual Relationship Manager

4. Meera Iyer

Role: Branch Manager

5. Rahul Menon

Role: Regional Head

6. Ananya Kapoor

Role: Head Office Executive

The active role must change:

- navigation
- homepage
- available actions
- metrics
- customer visibility
- workflow priorities

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7. REQUIRED DEMO CUSTOMERS

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Seed at least 30 realistic fictional customers.

Include these showcase customers:

CUSTOMER A:
Kumar Textiles Pvt Ltd

Type:
MSME

Location:
Chennai

Relationship Value:
₹48 lakh

Current Products:
- Current Account
- POS Solution
- Working Capital Facility

Signals:
- Turnover increased 31%
- Recent ZRT visit
- Expansion requirement
- One unresolved POS complaint

AI:
- Lead conversion probability: 87%
- Expansion finance propensity: 84%
- Churn risk: 24%

Next Best Action:
"Resolve open POS complaint, then schedule expansion finance consultation within 48 hours."

Reason codes:
- turnover growth
- explicit expansion need
- documents partially available
- unresolved service issue

CUSTOMER B:
Anita Sharma

Type:
Premium Individual

Relationship Value:

₹32 lakh

Signals:

- balance decreased 65%
- salary credit stopped
- two unresolved service issues
- no RM interaction for 76 days

AI:

- Churn risk: 82%

Next Best Action:

"Prioritize service recovery before any cross-sell discussion."

CUSTOMER C:

ABC Manufacturing

Type:

MSME

Signals:

- 15 recurring salary-like payments
- growing monthly transaction volume
- no salary account product

AI:

"Potential salary account opportunity for 15 employees."

Create additional customers across:

- Retail
- Premium
- MSME
- Emerging Business
- Dormant
- High Growth
- At Risk
- Digital First

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8. APPLICATION SHELL

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Create a premium application shell.

LEFT SIDEBAR:

Logo

YellowSense Customer 360

Navigation:

- Command Center
- Customer 360
- ZRT Field
- RM Workspace
- VRM Desk
- Leads
- Opportunities
- Meetings
- Complaints
- AI Copilot
- Analytics
- Audit Trail

Bottom:

- Active role
- Settings
- Help

TOP BAR:

- Global customer search
- Command palette trigger
- Region selector
- Notification bell
- AI Copilot shortcut
- User profile

Global search must find customers by:

- name
- customer number
- mobile
- business name

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9. PAGE 1 — EXECUTIVE COMMAND CENTER

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Route:

/executive

Build a premium enterprise dashboard.

Hero header:

"Good morning, Ananya"

"Here is where customer growth and execution need attention today."

Top KPIs:

- Relationship Value
- Active Customers
- High-Priority Opportunities
- Conversion Rate
- Business Mobilized
- Customers at Risk

Each KPI must show:

- current value
- trend
- comparison period
- tooltip definition

Required visuals:

A. Business Mobilization Trend

Line or area chart.

B. Lead Funnel

New

Qualified

Application

Approved

Converted

C. Regional Performance

Ranked branches/regions.

D. Opportunity Heatmap

Value vs Conversion Probability.

E. AI Intervention Queue

Examples:

- 18 high-value leads need action
- 7 complaints likely to breach SLA
- 12 customers show churn signals

- 23 dormant accounts have reactivation potential

F. Field-to-Conversion Funnel

Visits

Needs Detected

Leads

Qualified

Converted

G. "What Changed Today?" panel

Generate realistic operational insights.

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10. PAGE 2 — CUSTOMER 360

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Route:

/customers/[customerId]

This is the centerpiece of the POC.

Create a visually stunning customer profile.

HEADER:

Kumar Textiles Pvt Ltd

MSME • Chennai • Customer since 2021

Show:

- Relationship Value
- Segment
- Assigned RM
- Assigned VRM
- Sentiment
- Churn Risk
- Opportunity Potential

Primary actions:

- Schedule Meeting
- Create Lead
- Add Interaction
- Start Visit
- Open AI Brief

MAIN LAYOUT:

LEFT:

Customer identity and relationship card.

CENTER:

Relationship timeline.

RIGHT:

AI Intelligence panel.

Tabs:

1. Overview
2. Accounts & Products
3. Interactions
4. Visits
5. Leads
6. Meetings
7. Complaints
8. Documents
9. Consent
10. Audit

AI INTELLIGENCE PANEL:

Show:

NEXT BEST ACTION

"Resolve open POS complaint, then schedule expansion finance consultation."

Confidence:

87%

WHY:

- Turnover increased 31%
- Explicit expansion need captured
- Documents partially available
- One unresolved service issue

Recommended Owner:

Priya Nair

Recommended Timing:

Within 48 hours

Buttons:

- Accept Recommendation
- Modify
- Dismiss

When accepted:

- create a task
- write an audit event
- update UI
- show toast
- reflect in activity timeline

Also show:

- Product Propensity
- Churn Risk
- Lead Potential
- Relationship Sentiment

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11. PAGE 3 — ZRT FIELD COMMAND CENTER

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Route:

/zrt

Create a field operations dashboard.

Top section:

"Today's Priority Visit Plan"

Show:

- map-style panel
- numbered route pins
- total travel estimate
- priority opportunity value

Visit cards:

09:00

ABC Textiles

89% opportunity score

Reason: High growth signal

10:30

Kumar Enterprises

Follow-up overdue

Potential: ₹25L

12:00

XYZ Traders

Dormant account reactivation potential

Required actions:

- Start Visit
- Reschedule
- View Customer
- Navigate simulation

Add:

"AI Optimized Route"

Explain:

- opportunity score
- urgency
- distance
- customer availability

Do not pretend to use live maps unless an actual map provider is configured.

For the POC, use a polished abstract map visualization or open map integration.

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12. PAGE 4 — VISIT EXECUTION

=====

Route:

/zrt/visits/[visitId]

Create a mobile-inspired field workflow.

Steps:

1. Check In
2. Customer Verification
3. Profile Review
4. Need Assessment
5. Product Suggestions

6. Notes
7. Documents
8. Create Lead
9. Schedule Follow-up
10. Complete Visit

Show geo-verification as simulated.

Need assessment form:

- Working Capital
- Term Loan
- POS/QR
- Salary Accounts
- Insurance
- Other

After need assessment:

AI should generate product suggestions.

Example:

Working Capital Loan

Propensity: 84%

POS Upgrade

Propensity: 71%

Salary Accounts

Propensity: 68%

Each recommendation must show reason codes.

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13. OFFLINE-FIRST SIMULATION

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Create a visible network status control in development/demo mode.

States:

- Online
- Slow Network
- Offline

When Offline:

- allow visit notes
- allow need assessment
- allow lead draft
- queue mutations locally
- display pending sync count

Show:

OFFLINE MODE

4 records saved locally

2 documents queued

1 lead pending sync

When switched Online:

- simulate background synchronization
- update progress
- mark records synced
- show timestamps

Implement this meaningfully using IndexedDB or another browser-local persistence mechanism.

Do not merely display fake offline labels.

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14. PAGE 5 — RM WORKSPACE

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Route:

/rm

Build a high-value relationship intelligence cockpit.

Header:

"Good morning, Priya"

Subtext:

"8 customer actions require attention today."

Top KPIs:

- Portfolio Customers
- Relationship Value
- High Risk
- Cross-Sell Ready
- Follow-ups Due

LEFT:
Priority Action Queue.

Example:

01
Kumar Textiles
₹25L opportunity
87% conversion probability

WHY NOW:
- Asked about expansion
- Documents available
- Follow-up overdue

NEXT ACTION:
Call before 11:30 AM

02
Anita Sharma
82% churn risk

WHY:
- Balance down 65%
- Salary stopped
- unresolved complaints

NEXT ACTION:
Service recovery call

CENTER:
Portfolio Matrix.

Axes:
X = Growth Potential
Y = Relationship Value

Quadrants:
- Grow
- Retain
- Nurture
- Activate

Customers should be clickable.

RIGHT:

Selected Customer Snapshot.

Show:

- products
- relationship value
- sentiment
- open leads
- complaints
- next best action

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15. PAGE 6 — VRM DESK

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Route:

/vrm

Create a virtual engagement workspace.

Sections:

- A. Priority Contact Queue
- B. Customer Context Panel
- C. Interaction History
- D. Query/Complaint Context
- E. AI Suggested Conversation Preparation
- F. Post-Call Notes
- G. Follow-up Creation

When selecting a customer:

show:

- why the customer is prioritized
- recent interactions
- unresolved issues
- relevant opportunities

- communication consent
- preferred language
- recommended action

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16. PAGE 7 — LEAD & OPPORTUNITY PIPELINE

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Route:

/leads

Build both:

- Kanban pipeline
- advanced table view

Stages:

New

Contacted

Qualified

Documents Pending

Application

Approved

Converted

Lost

Lead card:

- customer
- product
- potential value
- owner
- AI conversion score
- next action
- aging
- source

Allow:

- drag between stages
- open detail drawer
- assign owner
- add follow-up
- convert lead
- mark lost with reason

When stage changes:

- persist change
- create audit event
- update analytics
- show timeline entry

Filters:

- region
- branch
- owner
- product
- score
- aging
- source

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17. PAGE 8 — MEETING INTELLIGENCE

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Route:

/meetings

Create a realistic meeting intelligence workflow.

Provide a sample fictional transcript.

Button:

"Generate Meeting Intelligence"

Output:

MEETING SUMMARY

Customer discussed:

- business expansion
- ₹25L funding requirement
- new warehouse plan

CONCERNS:

- interest rate
- processing time

ACTION ITEMS:

- send eligibility details
- collect GST statements

- follow up Friday

OPPORTUNITY:

MSME Expansion Finance

SENTIMENT:

Positive

Buttons:

- Create Lead
- Create Tasks
- Schedule Follow-up
- Save Summary

These buttons must work.

For POC:

Use deterministic/mock AI if no LLM API key exists.

If an LLM provider is configured:

use provider abstraction.

Never make the app unusable without an external AI key.

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18. PAGE 9 — COMPLAINT & QUERY INTELLIGENCE

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Routes:

/complaints

/queries

Create an operational service intelligence dashboard.

Complaint example:

"My EMI was deducted twice and nobody has resolved it."

AI output:

Intent:

Loan EMI Issue

Urgency:

High

Sentiment:

Negative

Suggested Team:

Loan Servicing

SLA Risk:

High

Recommended Priority:

P1

Required visuals:

- complaint volume trend
- category distribution
- SLA risk queue
- sentiment trend
- root-cause clusters

Allow:

- assign
- escalate
- resolve
- add note

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19. PAGE 10 — AI COPILOT

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Route:

/ai-copilot

Do not build a generic ChatGPT clone.

Build a context-aware banking employee copilot.

Suggested prompts:

- What should I focus on today?
- Summarize Kumar Textiles before my meeting.
- Which high-value leads are aging?
- Which customers show churn risk?
- Why is Anita Sharma marked high risk?

- Show cross-sell opportunities in my portfolio.

Response example:

Kumar Textiles has been a customer since 2021.

Current relationship:

- Current Account
- POS Solution
- Working Capital Facility

Recent signals:

- Turnover increased 31%
- Expansion need captured during ZRT visit
- POS complaint remains unresolved

Recommendation:

Resolve the POS issue before discussing expansion finance.

Potential opportunity:

₹25L

Every answer must show:

- source/context chips
- confidence where appropriate
- clear distinction between fact and recommendation

For POC:

Implement deterministic retrieval from seeded platform data.

Optional:

Add provider-based LLM generation.

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20. PAGE 11 — ANALYTICS & MIS

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Route:

/analytics

Create tabs:

1. Business Mobilization
2. Lead Conversion

3. ZRT Performance
4. RM Performance
5. Customer Risk
6. Cross-Sell
7. Service
8. AI Effectiveness

Required metrics:

ZRT:

- visits
- completed visits
- geo-verified visits
- needs detected
- leads created
- business mobilized

RM:

- portfolio value
- follow-up completion
- conversions
- cross-sell
- customer risk
- complaints resolved

AI:

- recommendations generated
- acceptance rate
- modified rate
- dismissal rate
- conversion after acceptance
- model confidence distribution

Add:

- date filters
- branch filters
- region filters
- export CSV
- realistic tooltips

Route:
/audit

Build an audit timeline.

Example:

10:42 AM
AI Recommendation Generated
Customer: Kumar Textiles
Model: LeadScore v2.4
Confidence: 87%

10:45 AM
Recommendation Accepted
Actor: Priya Nair

10:46 AM
Follow-up Task Created
Due: Tomorrow 11:30 AM

Show filters:

- actor
- entity
- action
- date
- AI model

Create an AI Decision Detail drawer:

Recommendation:
Prioritize RM follow-up

Confidence:
87%

Reason Codes:
- High engagement
- Documents available
- Expansion signal

Model:
LeadScore v2.4

Employee Decision:
Accepted

Timestamp:
10:45 AM

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22. AI IMPLEMENTATION STRATEGY

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Do not fake AI architecture.

Implement a clean abstraction.

Create interfaces/services for:

- A. Lead Scoring
- B. Churn Risk
- C. Product Propensity
- D. Next Best Action
- E. Sentiment
- F. Query Routing
- G. Meeting Summarization
- H. Copilot Retrieval

POC DEFAULT MODE:

Use deterministic explainable scoring.

Example lead score:

```
score =  
  engagement_weight  
+ explicit_need_weight  
+ document_readiness_weight  
+ relationship_weight  
- inactivity_penalty  
- unresolved_service_penalty
```

Normalize to 0–100.

Return:

```
{
```

```

"score": 87,
"band": "HIGH",
"reason_codes": [
  {
    "code": "EXPLICIT_EXPANSION_NEED",
    "impact": "positive",
    "contribution": 22
  },
  {
    "code": "DOCUMENTS_AVAILABLE",
    "impact": "positive",
    "contribution": 16
  },
  {
    "code": "OPEN_COMPLAINT",
    "impact": "negative",
    "contribution": -8
  }
],
"model_name": "poc-lead-score",
"model_version": "1.0"
}

```

Make every AI score explainable.

OPTIONAL PROVIDER MODE:

Support provider abstraction through environment variables.

Never hardcode secrets.

If provider unavailable:
fallback to deterministic POC mode.

```

=====
23. NEXT BEST ACTION ENGINE
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```

Implement a practical rules + score hybrid.

Example:

IF open severe complaint:
 prioritize service recovery

ELSE IF churn risk > 75:
prioritize retention outreach

ELSE IF high-value lead score > 80:
prioritize opportunity follow-up

ELSE IF product propensity > 75:
recommend eligible product discussion

ELSE IF follow-up overdue:
prioritize follow-up

Return:

- action
- priority
- confidence
- reason codes
- recommended owner
- recommended timing

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24. PRODUCT PROPENSITY

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Implement explainable POC propensity logic for:

- Working Capital Loan
- Term Loan
- Current Account
- POS / QR
- Salary Accounts
- Fixed Deposit
- Credit Card
- Insurance Referral

Example:

Salary Account propensity increases when:

- recurring salary-like payments exist
- employee count is known
- no salary product exists

Working Capital propensity increases when:

- MSME
- turnover growth
- working capital need captured
- transaction volume growth

Do not recommend ineligible products blindly.

Include a simple eligibility gate.

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25. END-TO-END GOLDEN DEMO JOURNEY

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This is mandatory.

Implement this exact demo:

STEP 1:

Login/switch role to Arjun Rao, ZRT Officer.

STEP 2:

Open Today's Visits.

STEP 3:

Select Kumar Textiles.

STEP 4:

Start Visit.

STEP 5:

Complete need assessment:

- Working Capital: Yes
- Expansion Finance: Yes
- POS issue: Yes

STEP 6:

AI generates:

- Expansion Finance propensity: 84%
- Lead conversion probability: 87%

STEP 7:

Create ₹25L lead.

STEP 8:

System recommends Priya Nair as owner.

Reason:

- MSME expertise
- branch proximity
- available capacity
- preferred language match

STEP 9:

Switch role to Priya Nair.

STEP 10:

The new lead appears in RM Priority Queue.

STEP 11:

Open Customer 360.

STEP 12:

See:

- ZRT visit
- need assessment
- lead
- open complaint
- AI recommendation

STEP 13:

Accept Next Best Action:

"Resolve POS complaint, then schedule expansion finance consultation."

STEP 14:

Create meeting.

STEP 15:

Open Meeting Intelligence.

STEP 16:

Generate summary and action items.

STEP 17:

Advance lead stage.

STEP 18:

Switch to Branch Manager.

STEP 19:

Dashboard reflects:

- visit
- lead
- opportunity value
- conversion funnel movement

This complete journey must work without manual database editing.

=====

26. SECOND GOLDEN DEMO JOURNEY

=====

Implement:

Anita Sharma

- churn risk 82%
- RM sees priority alert
- opens Customer 360
- reason codes visible
- open complaints visible
- Next Best Action says service recovery first
- RM accepts
- task created
- audit trail records decision

=====

27. INTERACTION DESIGN REQUIREMENTS

=====

Every important interaction must include:

- loading state
- success state
- error state
- empty state where relevant

Use:

- drawers for detail exploration
- modals only when necessary
- toasts for confirmation
- breadcrumbs on deep pages
- keyboard accessible controls

- visible focus states

Tables:

- sorting
- filtering
- pagination
- column visibility where useful

Charts:

- tooltips
- legends
- meaningful labels
- no meaningless decoration

=====

28. DEMO MODE

=====

Create a visible but discreet "Demo Mode" indicator.

Add:

"Reset Demo"

This should:

- restore seeded data
- restore lead stages
- restore AI decisions
- restore audit trail
- restore visit status

Create:

scripts/reset_demo.py

Also expose a safe development-only reset endpoint.

=====

29. DATA QUALITY

=====

Seed at least:

- 30 customers
- 50 accounts

- 40 product holdings
- 60 interactions
- 25 visits
- 35 leads
- 15 opportunities
- 20 meetings
- 18 complaints
- 20 queries
- 40 AI recommendations
- 100 audit events

Use deterministic seed values.

Use INR formatting:

₹48L

₹1.2Cr

Use Indian locations:

- Chennai
- Coimbatore
- Bengaluru
- Hyderabad
- Mumbai
- Pune

Use fictional names only.

=====

30. API REQUIREMENTS

=====

Create versioned REST endpoints.

Examples:

GET /api/v1/customers
GET /api/v1/customers/{id}
GET /api/v1/customers/{id}/timeline
GET /api/v1/customers/{id}/recommendations

GET /api/v1/visits
POST /api/v1/visits
POST /api/v1/visits/{id}/check-in
POST /api/v1/visits/{id}/complete

GET /api/v1/leads
POST /api/v1/leads
PATCH /api/v1/leads/{id}/stage
POST /api/v1/leads/{id}/assign

GET /api/v1/meetings
POST /api/v1/meetings
POST /api/v1/meetings/{id}/generate-intelligence

GET /api/v1/complaints
PATCH /api/v1/complaints/{id}

POST /api/v1/ai/lead-score
POST /api/v1/ai/churn-risk
POST /api/v1/ai/product-propensity
POST /api/v1/ai/next-best-action
POST /api/v1/ai/query-route
POST /api/v1/ai/meeting-summary

GET /api/v1/analytics/executive
GET /api/v1/analytics/zrt
GET /api/v1/analytics/rm

GET /api/v1/audit

Use:

- request schemas
- response schemas
- error schemas
- pagination
- filtering

31. SECURITY FOR POC

Implement sensible POC controls:

- JWT or secure session authentication
- password hashing
- role-based route authorization
- role-based API authorization
- masked account numbers

- no secrets committed
- .env.example
- CORS configuration
- input validation
- audit logging for sensitive actions

Do not claim production security certification.

=====

32. DOCUMENTATION

=====

Create:

README.md

Include:

- product overview
- screenshots placeholder section
- prerequisites
- local setup
- environment variables
- Docker setup
- seed instructions
- demo users
- golden demo journey
- test commands

ARCHITECTURE.md

Include:

- system context
- frontend architecture
- backend architecture
- AI abstraction
- data model
- integration adapter pattern
- POC vs production boundaries
- future event-driven evolution

docs/demo/DEMO_SCRIPT.md

Create a 7-minute bank pitch demo script.

docs/architecture/PRODUCTION_EVOLUTION.md

Explain how the POC can evolve toward:

- CDC
- Kafka
- schema registry
- durable workflow orchestration
- analytics store
- MLOps
- secure model serving
- hybrid deployment

Do not implement those heavy systems in the POC unless necessary.

=====

33. TESTING REQUIREMENTS

=====

Backend tests:

- customer retrieval
- lead creation
- stage transition
- visit completion
- AI score explanation
- NBA rule priority
- audit event creation

Frontend tests:

- customer 360 renders
- role switch changes navigation
- lead stage update
- AI recommendation acceptance
- offline queue behavior

Playwright E2E:

- Golden Demo Journey 1
- Golden Demo Journey 2

=====

34. DEFINITION OF DONE

=====

The project is complete only when:

1. It runs locally.
2. Docker Compose works.
3. Database migrations work.
4. Seed data loads.
5. Demo login works.
6. Role switching works.
7. Customer 360 works.
8. ZRT visit workflow works.
9. Need assessment works.
10. Lead creation works.
11. Smart assignment works.
12. RM priority queue updates.
13. AI recommendation acceptance works.
14. Audit trail updates.
15. Meeting intelligence works.
16. Analytics reflect workflow changes.
17. Offline queue simulation works meaningfully.
18. Reset Demo works.
19. Tests pass.
20. README contains exact setup commands.

=====

35. EXECUTION STRATEGY

=====

Do not attempt random file generation.

Work in this order:

PHASE 1

Analyze requirements and create:

- implementation plan
- architecture decisions
- route map
- data model
- component map

PHASE 2

Scaffold monorepo.

PHASE 3

Implement database models, migrations and deterministic seed data.

PHASE 4

Implement backend APIs.

PHASE 5

Implement application shell and design system.

PHASE 6

Implement Customer 360.

PHASE 7

Implement ZRT workflow.

PHASE 8

Implement RM/VRM workflow.

PHASE 9

Implement lead pipeline.

PHASE 10

Implement AI decision services.

PHASE 11

Implement meeting intelligence and copilot.

PHASE 12

Implement analytics and audit.

PHASE 13

Implement offline queue.

PHASE 14

Implement tests.

PHASE 15

Run:

- lint
- type check
- backend tests
- frontend tests
- E2E tests
- production build

Fix all critical failures.

=====

36. FINAL OUTPUT REQUIRED

=====

At completion, provide:

1. Final folder tree
2. Architecture summary
3. Exact local setup commands
4. Docker commands
5. Demo credentials
6. 7-minute demo sequence
7. Implemented features
8. POC limitations
9. Production evolution recommendations
10. Test results
11. Known issues, if any

IMPORTANT:

Do not stop after scaffolding.

Do not stop after creating a dashboard.

Do not leave primary pages as TODO.

Do not use lorem ipsum.

Do not create non-functional buttons for core flows.

Do not silently fake backend persistence.

Do not silently fake AI results without labeling the POC mode in technical documentation.

Do not expose internal implementation details unnecessarily in the customer-facing UI.

The final product must feel like a credible enterprise banking platform that YellowSense Technologies can demonstrate to a potential bank.

Begin now by:

1. producing the implementation plan,
2. creating the architecture decisions,
3. scaffolding the repository,
4. then implementing phase by phase without waiting for further confirmation.

Github details:

```
git remote add origin https://github.com/Vicky-cmd-run/UCO_yellowsense.git
```

```
git branch -M main
```

```
git push -u origin main
```